# MCRS LIGHTWEIGHT
# Master Engineering Baseline — Document 00

Scalable four-tower modular isolation system for light and specialty vehicles. Suspends the occupant cabin above the rolling chassis, isolates road shock, and provides COMFORT and SPORT modes. Powertrain-agnostic. Path A licensable module only.

*Always captured. Soft when you want comfort. Locked when you need control. Settles on passive support when power is gone.*

| 4 TOWERS | ±50 mm | COMFORT / SPORT | PATH A |
|---|---|---|---|
| Architecture | Macro stroke | Light profile modes | Buyer brings host |

| | |
|---|---|
| Document ID | MCRS-LW-DOC-00-R1.2 |
| Issue date | 2026-10-23 |
| Revision | R1.2 — expanded OEM baseline (~20-page target) |
| Product name | MCRS Lightweight (MCRS-LW) |
| Licensable unit | MCRS-MOD-001 — four-tower isolation module (Lightweight profile) |
| Legal Owner | Amirbahador Atai |
| Classification | T1 Confidential / NDA controlled |
| Evidence / TRL | L0 — TRL 2–3 — NOT CERTIFIED |
| Scope | Light / specialty vehicles — occupant cabin suspension |
| Out of scope | Heavyweight DOC-00; six-tower; crane/gantry; host powertrain |

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## 1. Document control and configuration

| Field | Controlled value |
|---|---|
| Document ID | MCRS-LW-DOC-00-R1.2 |
| Title | MCRS Lightweight — Master Engineering Baseline (Document 00) |
| Issue date | 2026-10-23 |
| Founder / Legal Owner | Amirbahador Atai (individual rights holder) |
| Licensable product | MCRS-MOD-001 four-tower isolation module, Lightweight profile |
| Status | Proposed baseline pending owner release and independent technical review |
| Evidence level | L0 — analytical and design-basis only |
| TRL | 2–3 — NOT CERTIFIED — no type-approval claim |
| Distribution | T1 NDA controlled |
| Independent Technical Reviewer | Due no later than earlier of 2026-12-22 and G1 closure |
| Predecessor | R1.0 / R1.1 — content retained and expanded |
| Companion (future) | MCRS Heavyweight DOC-00 — not in this release |

*Host powertrain is intentionally undefined. Buyer supplies ICE, hybrid or battery-electric architecture. Numerical values are stated once; downstream documents cite this DOC-00. Six-tower and Heavyweight profiles are configuration-excluded.*

## 2. Purpose, Path A model and scope boundary

MCRS Lightweight is a modular isolation product. Four capture towers support a rigid upper safety body — the occupant cabin. Road vibration and chassis torsion are attenuated before they reach occupants. In rapid manoeuvres the cabin can be mechanically locked to restore stiffness. On power loss the cabin settles onto passive hydraulic support and progressive stops.

Typical applications: specialty passenger vehicles, limited-series utility, ambulance / medical transport (comfort context only; no clinical claim), light protected mobility where continuous comfort and occasional high-stiffness manoeuvring are both required.

### 2.1 Path A — primary commercial path

| Element | Definition |
|---|---|
| What is licensed | Four-tower module design package: architecture, requirements, interface envelope, verification plan |
| What the buyer provides | Host chassis, powertrain, body adaptations, vehicle integration, finished-vehicle certification |
| Option exercise | Paid Option typically includes SIM-001 screening run and ICD-001 draft |
| Not included | Complete vehicle sale; traction battery / motor / ICE supply; type approval of finished vehicle |
| Territory / term / fee | Counsel-controlled — not stated in this engineering baseline |

### 2.2 Scope boundary

| In scope | Out of scope |
|---|---|
| Four-tower cabin isolation module | Six-tower variants |
| COMFORT and SPORT modes | Heavyweight single-mode (separate DOC-00) |
| Module mass, stroke, static floors, S4 | Host curb, range, pack kWh, fuel economy |
| 48 V module domain and comfort buffer | Host HV traction system design |
| ICD hard points and keep-outs | Industrial crane / gantry products |
| Module-level 48 V EMC | Whole-vehicle type approval |

### 2.3 Why powertrain is excluded

Locking ICE, hybrid and battery architectures into one module specification would force every licensee into one propulsion story. Path A is propulsion-agnostic so OEMs can adopt the isolation module on chassis they already build. Reference hosts, if used for simulation, are study tools only and are not part of the licensed deliverable.

## 3. Product definition — MCRS Lightweight

The product is a captured mechanical floating safety body (CMFSB) with four corner towers. Each tower combines a hydraulic macro-stage, progressive mechanical stops, a fine micro-stage and a positive SPORT lock face. The upper body remains geometrically captured in every operating state; isolation is provided only inside a controlled envelope.

| Attribute | Value | Status |
|---|---|---|
| Product name | MCRS Lightweight (MCRS-LW) | BASELINE |
| Module ID | MCRS-MOD-001 (Lightweight profile) | BASELINE |
| Tower count | 4 (one per corner) | BASELINE |
| Suspended object | Occupant cabin / upper safety cell | BASELINE |
| Macro stroke | ±50 mm hydraulic | TARGET |
| Micro stroke | ±7.5 mm fine stage | TARGET |
| Modes | COMFORT, SPORT, HOLD; S4 on power loss | BASELINE |
| Supply domain | 48 V class, galvanically isolated from host HV | BASELINE |
| Design life | ≥ 10 000 h operational (module) | TARGET |
| Control bandwidth | ≥ 20 Hz | TARGET |
| First structural mode | ≥ 60 Hz | TARGET |
| Mode transition time | ≤ 3 s | TARGET |
| Ambient (module continuous) | –20 °C to +50 °C | TARGET |

Design philosophy: Capture is permanent. Isolation is conditional. Lock is mechanical. Power-loss behaviour is passive. Software may improve comfort; it is not the sole barrier between the cabin and an uncontrolled free state.

# 4. Mass and load envelope (min / max suspended mass)

Licensing and structural sizing are governed by the mass of the **suspended upper body** (cabin structure + fixed interior + design occupant load), not by host GVW alone.

| Parameter | Minimum | Maximum (baseline) | Notes |
|---|---|---|---|
| Suspended mass (design) | ≈ 800 kg | ≈ 2 200 kg | Cabin + trim + design occupants |
| Per-tower class | Class S planning | Class L | Corner bill drives module mass |
| Module P50 (4 corners + shared) | ≈ 280 kg (S planning) | ≈ 393 kg (L) | Shared hydraulics/ECU in L |
| Host GVW (reference only) | ≈ 2.5 t | ≈ 4.5 t specialty | Buyer validates host structure |
| Above max suspended mass | — | Out of LW baseline | Heavyweight or new class |

## 4.1 Static support floors

| Case | Class S | Class L | Role |
|---|---|---|---|
| Equal-share continuous (1.25 × max share) | ≥ 4.5 kN / tower | ≥ 10.0 kN / tower | Fatigue / continuous |
| One-tower-loss STUDY (rigid-plate) | ≈ 10 kN / survivor | ≈ 22 kN / survivor | Cylinder & accumulator sizing |

*One-tower-loss values are STUDY: rigid plate, equal spring rates, CG near centroid or ≤100 mm forward. STR-001 shall confirm. Equal-share values remain the continuous rating.*

## 4.2 Why suspended mass governs

The module reacts only the suspended cabin. Chassis and unsprung masses stay on the host suspension. A light cabin on a heavy chassis remains inside the Lightweight envelope; a heavy armour cabin on a light chassis may not. The buyer declares suspended mass at Option exercise. ICD-001 requires the declared mass and CG band as interface data.

## 4.3 Occupant and payload convention

Design occupant load is included inside the suspended-mass maximum. Cargo inside the cabin counts toward suspended mass. Cargo on the chassis outside the cabin does not. Mis-declaration is an integration error owned by the buyer.

# 5. Architecture — four-tower layer stack

Four towers under a rigid cabin form a hyperstatic support (one redundant degree of freedom). Load sharing depends on stiffness matching and must be verified (REQ-LW-39).

| Layer (bottom → top) | Function | Primary elements |
|---|---|---|
| Host interface | Reaction path into chassis | Hard points, ICD keep-outs |
| Capture tower structure | Vertical guidance and force path | Guides, cylinder body, lock face |
| Macro stage | ±50 mm isolation travel | Hydraulic actuator + accumulator |
| Progressive stops | Energy absorption before hard bottoming | Defined load-deflection curve |
| Micro stage | ±7.5 mm fine motion | Fine actuator / gap control |
| Upper safety cell | Occupant cabin | Rigid frame, seams, levers |

In normal operation the tower set is the sole mechanical connection between rolling chassis and suspended cabin. No free levitation path exists in any mode including S4.

## 5.1 Corner functional block

Each corner is a self-contained vertical unit: structural guide, hydraulic macro cylinder, port-mounted burst protection, progressive stop stack, fine stage, position sensing and SPORT lock face. Corners are intended to be serviceable as units.

## 5.2 Cabin rigidity assumption

Load-share and one-tower-loss STUDY models treat the upper cell as rigid relative to tower compliance. If a buyer cabin is significantly flexible, SIM-001 must be re-run with actual cabin stiffness.

# 6. Operating modes — COMFORT and SPORT

| Mode | Intent | Actuation state |
|---|---|---|
| COMFORT | Maximise isolation within the stroke envelope | Macro + micro active; lock disengaged |
| SPORT | Restore high roll/pitch stiffness for manoeuvres | Lock faces engaged; micro may trim |
| HOLD | Hold last commanded set-point | Valves held; sensors live |
| S4 | Power-loss or critical-fault safe state | Passive residual pressure + stops; no active power |

Mode transition time target ≤ 3 s. Host requests are advisory; module safety invariants override. Lateral-g interlock inhibits mode changes that would unload capture under high lateral acceleration.

## 6.1 COMFORT

Macro stage tracks ride-height set-point while rejecting road disturbance within bandwidth limits. Micro stage trims residual motion. Approaching stops triggers progressive stiffening and, if needed, transition toward HOLD. COMFORT is the default after successful power-up self-test unless the host requests otherwise and interlocks allow it.

## 6.2 SPORT

Lock faces engage so cabin roll and pitch stiffness rise to mechanical lock stiffness. Intended for rapid directional changes where isolation would reduce handling authority. Exit requires lock status confirmation before COMFORT is re-enabled. Disagreeing lock status means the module shall not claim SPORT.

## 6.3 HOLD

HOLD freezes the last valid set-point when sensor disagreement, degraded actuation or host request requires a stable intermediate state. HOLD is not a substitute for S4; loss of supply still forces S4.

# 7. Safety invariants and power-loss behaviour (S4)

| ID | Invariant |
|---|---|
| S0 | Upper body remains geometrically captured in every operating state — no free separation path |
| S1 | Progressive stops engage before hard structural bottoming |
| S2 | SPORT lock is positive and inspectable; locked state does not require continuous electrical power |
| S3 | Lateral-g interlock prevents capture-unloading transitions |
| S4 | On loss of module supply or critical fault, cabin settles on passive support + stops |
| S5 | Hose-burst protection at each cylinder port (velocity fuse or equivalent) limits outflow on the passive path |
| S6 | Sensor disagreement forces HOLD or S4 — never an open-loop force command |
| S7 | Manual rescue two-step mechanical procedure remains available without electrical power |
| S8 | Module EMC design shall not allow host fields to corrupt lock or isolation state |

## 7.1 Power-loss sequence (REQ-LW-07)

On loss of 48 V or critical fault the module enters S4. Valves move to passive-support position. Residual accumulator pressure and progressive stops carry the cabin. Comfort buffer is not credited in the S4 safety case. Velocity fuses remain in the passive hydraulic path. Manual rescue remains a two-step mechanical procedure independent of electrical state.

## 7.2 Manual rescue (summary)

Step 1: confirm vehicle stationary and secured. Step 2: apply the published mechanical release sequence for occupant egress. Full procedure is an operations document; it shall not require live module software.

## 7.3 Finished-vehicle functional safety

This baseline does not claim ISO 26262 ASIL ratings. Module invariants are written so a buyer safety case can allocate them. Any ASIL claim is a finished-vehicle activity owned by the licensee after integration evidence exists.

# 8. Structural principles and one-tower-loss STUDY

Continuous operation uses equal-share floors. Cylinder, accumulator and corner structure sizing is driven by the one-tower-loss STUDY case.

| Metric | Class S | Class L |
|---|---|---|
| Equal-share floor per tower | ≥ 4.5 kN | ≥ 10.0 kN |
| One-tower-loss floor per survivor (STUDY) | ≈ 10 kN | ≈ 22 kN |
| Governing use | Continuous / fatigue | Burst / residual three-tower cell |

*Assumptions: rigid upper cell, equal vertical stiffness, CG at centroid or ≤100 mm forward, hose-burst not stacked with tower loss. STR-001 confirms.*

## 8.1 Hyperstatic framing

Four supports on a rigid cell are statically indeterminate (one redundant degree). Stiffness matching, installation tolerance and thermal distortion affect load share (REQ-LW-39).

## 8.2 Stops and horizontal path

Progressive stops define the energy curve before hard bottoming. Horizontal capture-frame loads are bounded under REQ-LW-34. Stop efficiency below design assumption grows stack height (R-LW-07).

## 8.3 Screening loads

Vertical and horizontal screening cases for the capture frame are defined for analysis in STR-001. They are magnitude screens for structural members, not public-road performance claims. Buyer-specific crash or blast loads are outside this baseline unless added by controlled deviation.

# 9. Hydraulics, auxiliary energy and EMC

## 9.1 Hydraulic topology

A rigid cabin has three degrees of freedom in the vertical/roll/pitch plane that must be restrained. Three independent pressure circuits are required for static determinacy and passive roll restraint under S4 (example: front pair, left rear, right rear). Fewer than three circuits require explicit passive roll restraint and are not baseline. Each cylinder port shall include a velocity fuse or equivalent (S5 / REQ-LW-38).

| Circuit example | Corners served | Role under S4 |
|---|---|---|
| Front pair | FL + FR | Pitch / heave share |
| Left rear | RL | Roll restraint left |
| Right rear | RR | Roll restraint right |

## 9.2 Auxiliary energy (comfort only)

Peak fine-stage power at design motor constant can exceed continuous rating of a small DC-DC converter. A local 48 V buffer is mandatory for comfort duty. S4 does not rely on this buffer. Final energy from measured duty cycle.

| Relative Km | Relative peak power | Implication |
|---|---|---|
| Baseline (high Km) | 1.0 × | Buffer sized to DC-DC deficit |
| Km –30 % | ≈ 2.0 × | Larger buffer / thermal redesign |
| Much lower Km | ≈ 2.5–4 × | Duty limit or actuator change |

*Peak power scales ≈1/Km² for fixed force. ACT-001 characterises the actuator.*

## 9.3 Fluid and contamination

Fluid specification, filtration and water content limits are HYD-001 items. Contamination control is a durability and valve-reliability issue, not a substitute for S4.

## 9.4 EMC (module domain)

The 48 V module domain shall be designed for immunity to the host electromagnetic environment. Finished-vehicle EMC remains the buyer's responsibility (REQ-LW-37, EMC-001).

# 10. Packaging, hard points and interface control

Module vertical space claim after levers, stops and free play is on the order of 170–190 mm depending on host class (TARGET). Hard points, keep-outs, service access and connectors are controlled in ICD-001. Buyer adapts the chassis.

| Interface class | Owner | Document |
|---|---|---|
| Structural hard points and loads | Module defines envelope; host implements | ICD-001 |
| Hydraulic connections / fluid spec | Module | HYD-001 / ICD-001 |
| 48 V power and discrete signals | Module | ICD-001 |
| Mode request / status to host controller | Shared protocol | ICD-001 |
| Host traction / charging / ICE systems | Buyer only | Outside module scope |

## 10.1 Installation principles

Towers installed with controlled parallelism and height tolerance so static load share stays within verified band. ICD-001 states datum features, torque and inspection checks. Field adjustment shall not defeat capture geometry.

## 10.2 Service and access

Each corner shall allow removal of the serviceable unit without cutting host structure. Hydraulic and electrical disconnects are designed for workshop replacement. Roadside internal repair is not a design goal.

# 11. Sensing, diagnostics and control constraints

Module sensing supports height/gap, lock status, pressure (or force proxy) and lateral acceleration for interlock. Exact sensor technology is not frozen at L0; disagreement policy is frozen.

| Signal class | Use | Disagreement policy |
|---|---|---|
| Corner height / gap | Ride-height control, stop approach | HOLD or S4 if unresolved |
| Lock status | SPORT confirmation | Do not claim SPORT if disagree |
| Pressure / force proxy | Support monitoring, S4 health | Degrade to passive priority |
| Lateral-g | Mode interlock | Inhibit unload transitions |
| Supply voltage | S4 entry | Immediate S4 on critical loss |

## 11.1 Control constraints

Detailed control laws live in the control specification. This DOC-00 constrains them: bandwidth ≥ 20 Hz, first structural mode ≥ 60 Hz, mode time ≤ 3 s, no open-loop force under sensor disagreement, host commands advisory only.

## 11.2 Logging and service data

Minimum service data (fault codes, last mode, lock status, supply state) shall be readable via ICD-defined access. Full telemetry definition is ICD/control scope.

## 12. Environmental and durability envelope

Module-level environmental targets for design. Finished-vehicle climate and sealing remain buyer responsibilities beyond the module keep-out.

| Parameter | Target / limit | Notes |
|---|---|---|
| Continuous ambient | –20 °C to +50 °C | Module continuous operation |
| Survival ambient (non-operating) | –30 °C to +65 °C STUDY | Storage / soak |
| Design life | ≥ 10 000 h operational | Module |
| Dust / water | ICD keep-out + connector class | Not a full IP rating claim at L0 |
| Shock / vibration (module shipping) | STR-001 screen | Separate from road isolation duty |
| Corrosion protection | Material / coating bill | Manufacturing specification |

*Extreme climate validation is an open item. Buyers operating outside the continuous band must declare the environment at Option exercise.*

## 13. Requirements register (REQ-LW)

| ID | Requirement (summary) | Verification |
|---|---|---|
| REQ-LW-01 | Cabin remains captured in all operating states | Geometry / ICD-001 |
| REQ-LW-07 | Power loss → S4; buffer not credited for safety | Power-loss test |
| REQ-LW-18 | Mode arbitration not bypassable by host domain controller | HIL G3 |
| REQ-LW-19 | Sensor disagreement forces HOLD or S4 | Diagnostics policy |
| REQ-LW-22 | Leveling authority within macro stroke under design load | Bench G2 |
| REQ-LW-23 | Hydraulic safety consistent with ISO 4413 principles | HYD-001 |
| REQ-LW-34 | Horizontal capture-frame load bound | STR-001 |
| REQ-LW-37 | Module EMC immunity in stated environment | EMC-001 |
| REQ-LW-38 | Velocity fuse (or equivalent) at each cylinder port | HYD-001 |
| REQ-LW-39 | Load-share verification for four-tower hyperstatic cell | SIM-001 / G1 |

## 14. Verification gates and evidence packages

| Gate | Evidence package | Closes |
|---|---|---|
| G1 | SIM-001 screening + ICD-001 draft | Option exercise; load-share scheme |
| G2 | Tower bench, hydraulic failure test, actuator characterisation | STR / HYD / ACT |
| G3 | HIL mode transitions + power-loss to S4 | Control and S4 behaviour |

*No external safety-performance credit before independent technical review and gate closures. Finished-vehicle certification remains with the buyer.*

| Package | Purpose |
|---|---|
| SIM-001 | Multi-body / load-share screening under design mass envelope |
| ICD-001 | Hard points, connectors, signals, keep-outs |
| STR-001 | Structural floors, stops, one-tower-loss confirmation |
| ACT-001 | Actuator force / Km characterisation |
| HYD-001 | Circuit topology, accumulators, velocity fuse, ISO 4413 alignment |
| EMC-001 | Module-domain immunity plan |
| LCK-001 | SPORT lock geometry and power-loss behaviour |

## 15. Risks, assumptions and open items

| ID | Risk | S | P | Mitigation |
|---|---|---|---|---|
| R-LW-01 | Uneven load share on four-tower cell | 3 | 3 | REQ-LW-39, SIM-001 |
| R-LW-02 | Lock logic ambiguity on power loss | 3 | 4 | LCK-001 / ICD-001 |
| R-LW-03 | Hose burst on passive path | 5 | 2 | Velocity fuse REQ-LW-38 |
| R-LW-04 | Actuator constant below assumption | 3 | 3 | ACT-001 sensitivity |
| R-LW-05 | Class S corner mass distribution unproven | 3 | 4 | Detailed bill; owner decision |
| R-LW-06 | Host EMC environment worse than assumed | 3 | 3 | EMC-001; buyer host data |
| R-LW-07 | Stop efficiency below design assumption | 3 | 3 | STR-001; stack growth |
| R-LW-08 | Cabin flexibility invalidates rigid-cell STUDY | 3 | 3 | SIM-001 with buyer stiffness |

Assumptions: suspended mass within Section 4; equal spring rates for STUDY floors; adequate stop efficiency; chassis can react module loads; 48 V available when isolation requested; cabin stiffness adequate or declared for SIM-001.

Open items: comfort-buffer energy; measured Km; Class S detailed bill; independent reviewer; ICD connector table; STR-001 one-tower-loss confirmation; lateral-g thresholds; extreme ambient validation.

## 16. Work packages and lightweight document tree

Path A work uses module-primary packages in Section 14. Host vehicle documents are buyer-owned. Six-tower and Heavyweight profiles are excluded until their own DOC-00 is released.

| Tree item (indicative) | Role |
|---|---|
| DOC-00 (this document) | Controlling baseline |
| ICD-001 | Interface control |
| STR-001 / HYD-001 / ACT-001 | Structural, hydraulic, actuator evidence |
| EMC-001 / LCK-001 | EMC and lock evidence |
| Control specification + HIL plan | Modes, interlocks, S4 timing |
| Ops / rescue procedure | Manual rescue two-step |

## 17. Commercial Path A option outline

Engineering description of a typical Path A Option. Commercial terms (price, territory, exclusivity) are counsel-controlled and not stated here.

| Option element | Engineering content |
|---|---|
| Baseline access | This DOC-00 under NDA |
| SIM-001 run | Screening under buyer-declared suspended mass and CG band |
| ICD-001 draft | Hard points, connectors, signals adapted to buyer chassis datums |
| Open-item list | Which STUDY items must close before production intent |
| Not in Option | Tooling, prototype hardware, type approval, host powertrain design |

*Successful Option exercise does not by itself grant production licence rights; those follow the Path A agreement under counsel.*

## 18. IP, claims, release and glossary

Legal owner: Amirbahador Atai. Licensable product: MCRS-MOD-001 Lightweight profile. No certification or type-approval claim is made. Release for external safety-performance credit requires owner sign-off, independent technical review, and counsel review of Path A terms.

| Role | Name | Date | Signature |
|---|---|---|---|
| Owner release | Amirbahador Atai | To be completed | To be completed |
| Independent Technical Reviewer | To be appointed | To be completed | To be completed |
| Counsel (Path A terms) | To be appointed | To be completed | To be completed |

### 18.1 Glossary

| Term | Definition |
|---|---|
| MCRS Lightweight / MCRS-LW | Four-tower cabin isolation module product (this DOC-00) |
| MCRS-MOD-001 | Licensable four-tower module identifier |
| CMFSB | Captured Mechanical Floating Safety Body — upper cell always captured |
| COMFORT / SPORT / S4 | Isolation mode / mechanically locked mode / power-loss passive state |
| Suspended mass | Mass of cabin + fixed content under design load — governs sizing |
| Path A | Licence of the module; buyer supplies host and powertrain |
| Class S / L | Corner mass and load class |
| ICD-001 | Interface control document |

## Appendix A. Arithmetic verification

| Check | Basis | Result | Status |
|---|---|---|---|
| Equal-share floor Class S | 1.25 × max per-tower share | ≥ 4.5 kN | OK |
| Equal-share floor Class L | 1.25 × max per-tower share | ≥ 10.0 kN | OK |
| One-tower-loss Class S (STUDY) | ≈2.1 × (W/4) | ≈ 10 kN | STUDY |
| One-tower-loss Class L (STUDY) | ≈2.2 × (W/4) | ≈ 22 kN | STUDY |
| Module P50 Class L | 4 × 94 kg + shared allocation | ≈ 393 kg | OK |
| Module P50 Class S planning | 4 × 70 kg | ≈ 280 kg | OPEN distribution |
| Macro stroke total | (+50) – (–50) mm | 100 mm | OK |
| Micro stroke total | (+7.5) – (–7.5) mm | 15 mm | OK |

## Appendix B. Architecture comparison notes

Relative to passive cabin mounts or seat-only suspension, MCRS Lightweight isolates the entire cabin, offers a positive mechanical lock for manoeuvres, and defines an explicit power-loss passive state. Relative to fully active body control integrated by a single OEM, it is offered as a discrete licensable module so multiple host manufacturers can adopt the same isolation architecture without a prescribed powertrain. This appendix does not claim measured superiority on any public test cycle.

## Appendix C. Mode transition summary

| From \ To | COMFORT | SPORT | HOLD | S4 |
|---|---|---|---|---|
| COMFORT | — | Lock engage sequence | Freeze valves | Power-loss / critical fault |
| SPORT | Lock release + confirm | — | Freeze valves | Power-loss / critical fault |
| HOLD | Resume isolation | Lock engage sequence | — | Power-loss / critical fault |
| S4 | Power restore + checks | Power restore + checks | Power restore | — |

*All transitions that affect capture are interlocked. Timing and hysteresis are control-specification items.*

## Appendix D. Failure-case screening list

| ID | Failure mode | Immediate effect | Module response |
|---|---|---|---|
| F-01 | Loss of capture geometry | Free relative motion risk | Design exclusion (S0) |
| F-02 | Loss of passive pressure | Cabin on stops only | Accumulators; velocity fuse |
| F-03 | Reserved | — | Reserved for configuration control |
| F-04 | SPORT lock fails to engage | Reduced stiffness when requested | Inhibit SPORT claim |
| F-05 | SPORT lock fails to release | Stuck high stiffness | Manual rescue |
| F-06 | Hose burst on passive path | Loss of residual pressure | Port velocity fuse (S5) |
| F-07 | Sensor disagreement | Conflicting height/force data | HOLD or S4 (S6) |
| F-08 | Actuator below Km assumption | Thermal / force shortfall | ACT-001; duty limit |
| F-09 | One tower lost | Overload on survivors | STUDY floors; STR-001 |
| F-10 | EMC corruption of lock state | Uncommanded mode change risk | REQ-LW-37; EMC-001 |

## Appendix E. Corner mass planning bill

| Element (planning) | Class S (kg) | Class L (kg) | Notes |
|---|---|---|---|
| Structural guides / housing | TBD in bill | TBD in bill | STR-001 detail |
| Hydraulic cylinder + fluid share | TBD | TBD | HYD-001 |
| Stops + lock face | TBD | TBD | LCK-001 |
| Fine stage + sensors | TBD | TBD | ACT-001 |
| Corner total (planning) | 70 | 94 | P50 planning |
| Four corners | 280 | 376 | Before shared items |
| Shared hydraulics / ECU allocation | incl. in 280 planning | ≈ 17 | Module-level share |
| Module P50 total | ≈ 280 (planning) | ≈ 393 | Section 4 |

*Detailed Class S bill is OPEN. Contract sizing sensitive to Class S shall present both 70 kg planning and sensitivity at 84–94 kg.*

## Appendix F. Vertical stack contribution

| Element | Contribution (mm) | Notes |
|---|---|---|
| Tower structure / guides | Design-dependent | ICD-001 / STR-001 |
| Macro travel | 100 (±50) | Hydraulic stroke |
| Micro travel | 15 (±7.5) | Fine stage |
| Stop stroke allowance | Design-dependent | Progressive stops |
| Free play / installation tolerance | Design-dependent | Must be declared |
| Total space claim target | 170 (lighter) / 190 (heavier cabin class) | After all contributors |

## Appendix G. Buyer declaration checklist (Option)

Information the buyer should provide at Path A Option exercise so that SIM-001 and ICD-001 draft are meaningful.

| Item | Why required |
|---|---|
| Declared suspended mass and CG band | Selects Class S/L and drives STUDY floors |
| Cabin plan dimensions / hard-point preference | ICD geometry |
| Available vertical package height | Confirms 170–190 mm claim feasibility |
| 48 V supply capability and location | Module power interface |
| Host domain-controller interface preference | Mode request / status protocol |
| Operating ambient if outside –20…+50 °C | Environmental open item |
| Cabin stiffness class (rigid / flexible) | SIM-001 modelling choice |

*End of MCRS-LW-DOC-00-R1.2 — MCRS Lightweight Master Engineering Baseline. Legal Owner: Amirbahador Atai. NOT CERTIFIED. TRL 2–3. T1 NDA controlled. Powertrain-agnostic four-tower cabin isolation module — Path A licensable product. Heavyweight profile and six-tower variants are outside this baseline.*